

Chapter 04

Job Order Costing

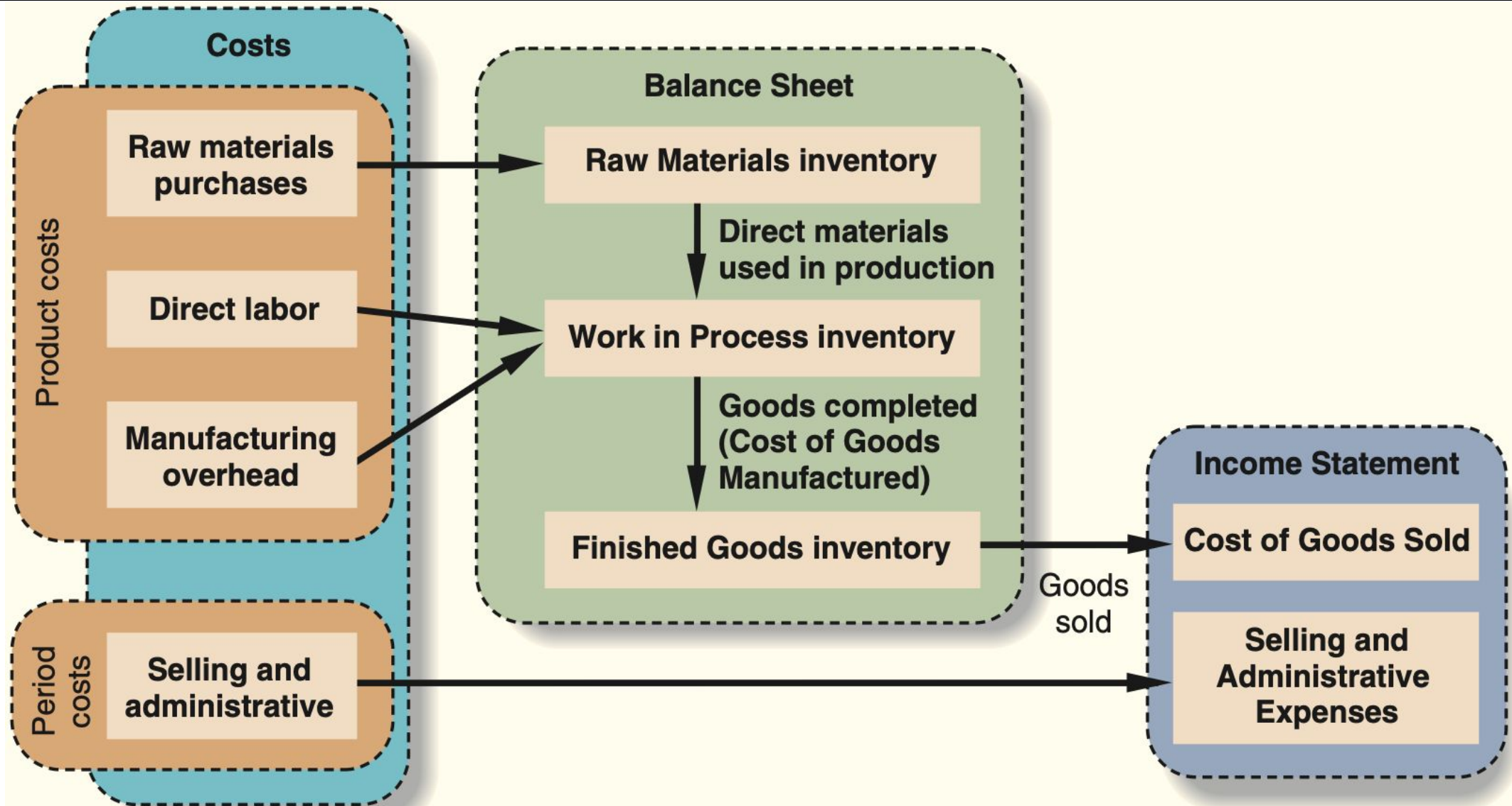
Double Entry System

Type	Increase	Decrease
Asset	Debit	Credit
Liability	Credit	Debit
Capital	Credit	Debit
Income	Credit	Debit
Expense	Debit	Credit

What is Job Costing?

Job Costing can be defined as “the category of basic costing methods which is applicable where the work consists of separate contracts, jobs or batches, each of which is authorised by specific order or contract.

Job-Order Costing—The Flow of Costs



Measuring Direct Materials Cost

- A **bill of materials** is a document that **lists** the type and quantity of each type of direct material needed to complete a unit of product.
- The **materials requisition form** is a document that **specifies** the type and quantity of materials to be drawn from the storeroom and identifies the job that will be charged for the cost of the materials.
- A **job cost sheet** records the materials, labor, and manufacturing overhead costs charged to that job.

JOB COST SHEET



Job Number	2B47	Date Initiated	March 2
Department	Milling	Date Completed	
Item	Special order coupling	Units Completed	
For Stock			

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
14873	\$660	843	5	\$45			





Cost Summary		Units Shipped	
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Measuring Direct Labor Cost

- Direct labor consists of labor charges that are easily traced to a particular job. Labor charges that cannot be easily traced directly to any job are treated as part of manufacturing overhead.

Computing Predetermined Overhead Rates

1. Manufacturing overhead is an indirect cost. This means that it is either impossible or difficult to trace these costs to a particular product or job.
2. Manufacturing overhead consists of many different types of costs ranging from the grease used in machines to the annual salary of the production manager. Some of these costs are variable overhead costs because they vary in direct proportion to changes in the level of production (e.g., indirect materials, supplies, and power) and some are fixed overhead costs because they remain constant as the level of production fluctuates (e.g., heat and light, property taxes, and insurance).
3. Because of the fixed costs in manufacturing overhead, total manufacturing overhead costs tend to remain relatively constant from one period to the next even though the number of units produced can fluctuate widely. Consequently, the average cost per unit will vary from one period to the next.

Computing Predetermined Overhead Rates

An allocation base is a measure such as direct labor-hours (DLH) or machine-hours (MH) that is used to assign overhead costs to products and services. The most widely used allocation bases in manufacturing are direct labor-hours, direct labor cost, machine-hours, and (where a company has only a single product) units of product.

$$\text{Predetermined overhead rate} = \frac{\text{Estimated total manufacturing overhead cost}}{\text{Estimated total amount of the allocation base}}$$

Computing Predetermined Overhead Rates

Time Ticket No.

843

Date

March 3

Employee

Mary Holden

Station

4

	Started	Ended	Time Completed	Rate	Amount	Job Number
	7:00	12:00	5.0	\$9	\$45	2B47
	12:30	2:30	2.0	9	18	2B50
	2:30	3:30	1.0	9	9	Maintenance
	Totals		8.0		\$72	

Applying Manufacturing Overhead

$$\text{Overhead applied to a particular job} = \text{Predetermined overhead rate} \times \text{Amount of the allocation base incurred by the job}$$

JOB COST SHEET

Job Number 2B47

Date Initiated March 2

Department Milling

Date Completed March 8

Item Special order coupling

For Stock

Units Completed 2

Direct Materials		Direct Labor			Manufacturing Overhead		
Req. No.	Amount	Ticket	Hours	Amount	Hours	Rate	Amount
14873	\$ 660	843	5	\$ 45	27	\$8/DLH	\$216
14875	506	846	8	60			
14912	238	850	4	21			
	\$1,404	851	10	54			
			27	\$180			

Cost Summary		Units Shipped		
		Date	Number	Balance
Direct Materials	\$ 1,404	March 8	—	2
Direct Labor	\$ 180			
Manufacturing Overhead	\$ 216			
Total Product Cost	\$ 1,800			
Unit Product Cost	\$ 900*			

*\$1,800 ÷ 2 units = \$900 per unit.

Choice of an Allocation Base for Overhead Cost

Computation of Unit Costs

Job Costing Exercise

we will consider a single month's activity at Ruger Corporation, a producer of gold and silver commemorative medallions. Ruger Corporation has two jobs in process during April, the first month of its fiscal year. Job A, a special minting of 1,000 gold medallions commemorating the invention of motion pictures, was started during March. By the end of March, \$30,000 in manufacturing costs had been recorded for the job. Job B, an order for 10,000 silver medallions commemorating the fall of the Berlin Wall, was started in April.

The Purchase and Issue of Materials

On April 1, Ruger Corporation had \$7,000 in raw materials on hand. During the month, the company purchased on account an additional \$60,000 in raw materials. The purchase is recorded in journal entry (1) below:

	(1)	Dr	CR
Raw Materials			60,000
Accounts Payable		60,000	

The Purchase and Issue of Materials

During April, \$52,000 in raw materials were requisitioned from the storeroom for use in production. These raw materials included \$50,000 of direct and \$2,000 of indirect materials. Entry (2) records issuing the materials to the production departments.

(2) DR CR

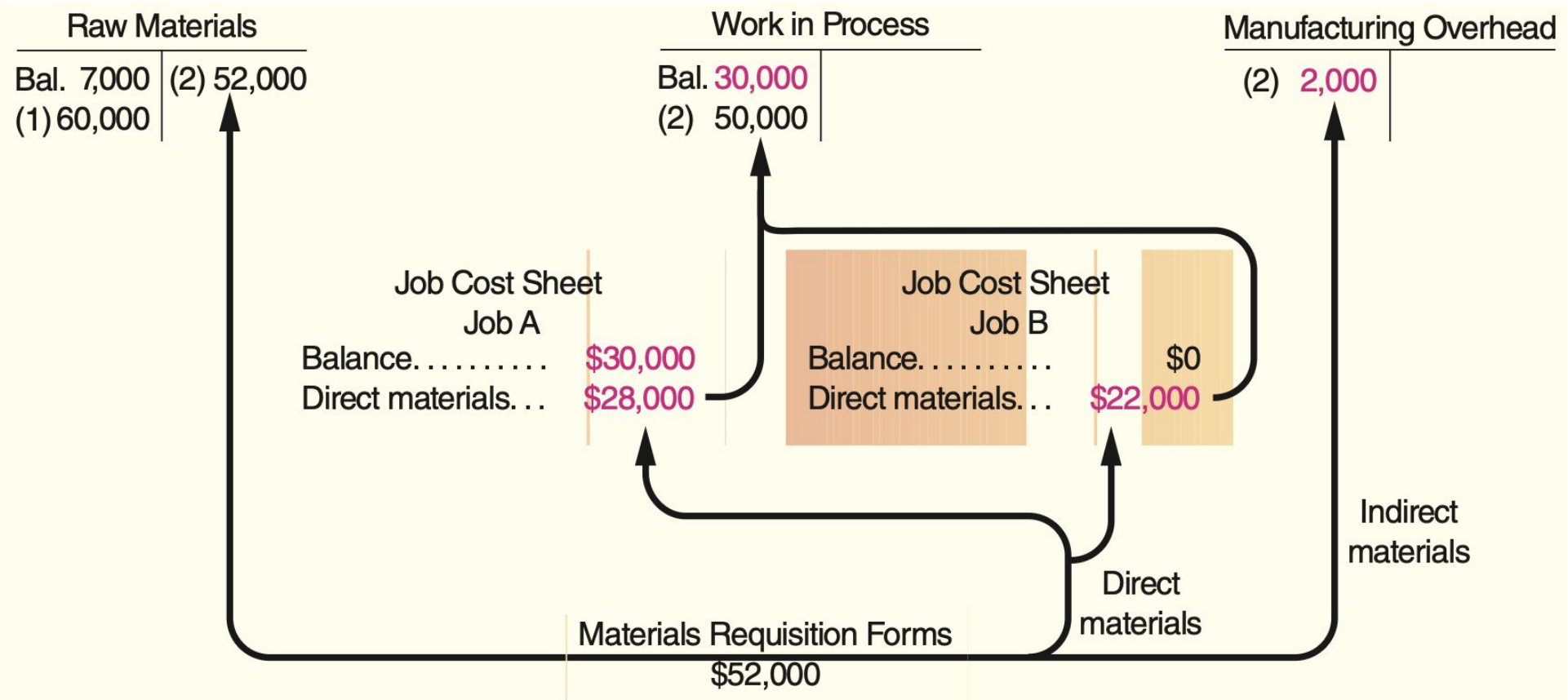
Work in Process 50,000

Manufacturing Overhead 2,000

Raw Materials 52,000

where \$28,000 of the \$50,000 in direct materials is charged to Job A's cost sheet and the remaining \$22,000 is charged to Job B's cost sheet.

Raw Materials Cost Flows



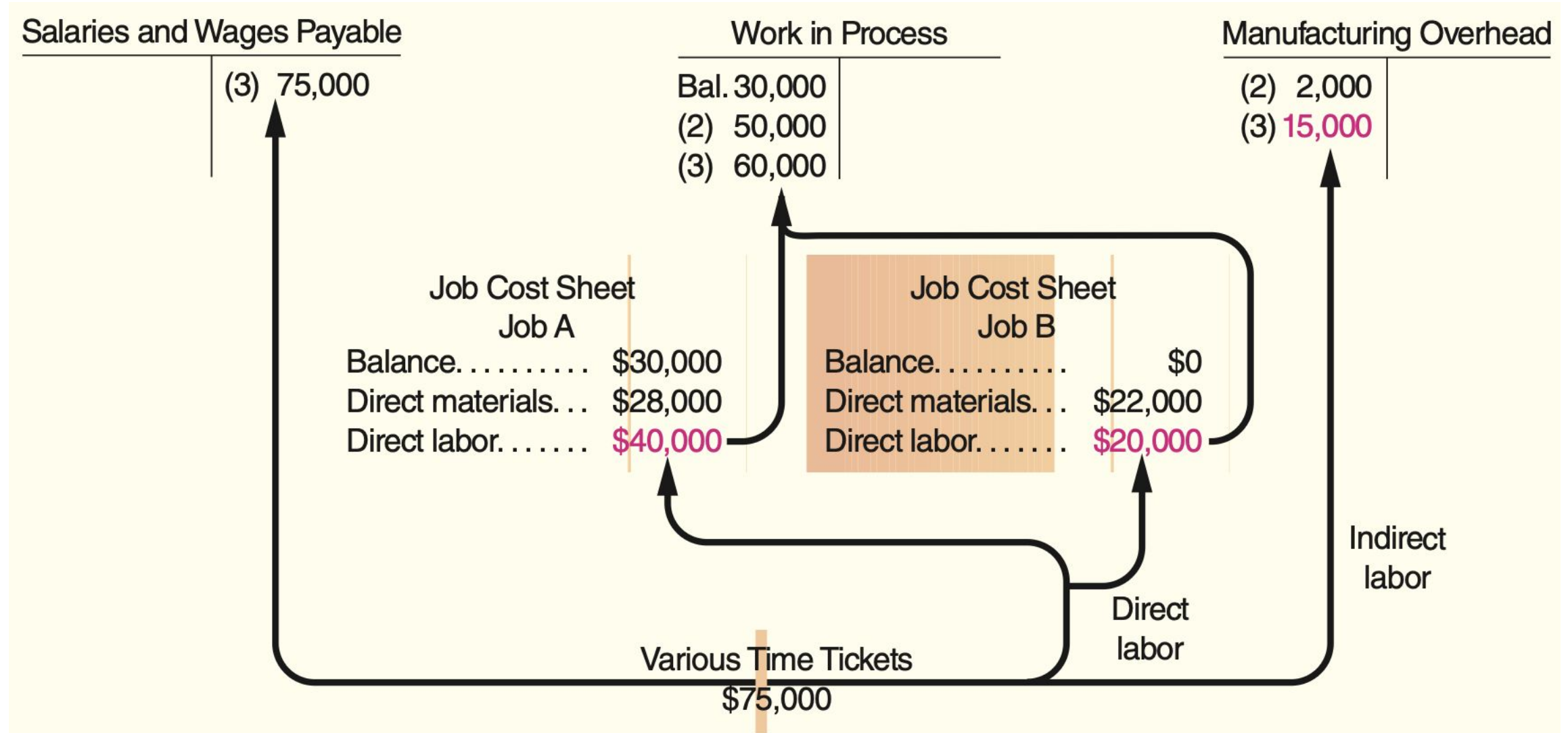
Labor Cost

In April, the employee time tickets included \$60,000 recorded for direct labor and \$15,000 for indirect labor. The following entry summarizes these costs:

(3)	DR	CR
Work in Process	60,000	
Manufacturing Overhead	15,000	
Salaries and Wages Payable		75,000

During April, \$40,000 of direct labor cost was charged to Job A and the remaining \$20,000 was charged to Job B.

Labor Cost Flows



Manufacturing Overhead Costs

To illustrate, assume that Ruger Corporation incurred the following general factory costs during April:

Utilities (heat, water, and power)	\$21,000
Rent on factory equipment	16,000
Miscellaneous factory overhead costs	<u>3,000</u>
Total	<u><u>\$40,000</u></u>

The following entry records the incurrence of these costs:

(4)	DR	CR
Manufacturing Overhead	40,000	
Accounts Payable *		40,000

*Accounts such as Cash may also be credited

Manufacturing Overhead Costs

In addition, assume that during April, Ruger Corporation recognized \$13,000 in accrued property taxes and that \$7,000 in prepaid insurance expired on factory buildings and equipment. The following entry records these items:

(5)	DR	CR
Manufacturing Overhead		20,000
Property Taxes Payable		13,000
Prepaid Insurance		7,000

Finally, assume that the company recognized \$18,000 in depreciation on factory equipment during April. The following entry records the accrual of this depreciation:

(6)	DR	CR
Manufacturing Overhead		18,000
Accumulated Depreciation		18,000

Applying Manufacturing Overhead

Assume that Ruger Corporation’s predetermined overhead rate is \$6 per machine-hour. Also assume that during April, 10,000 machine-hours were worked on Job A and 5,000 machine-hours were worked on Job B (a total of 15,000 machine-hours). Thus, \$90,000 in overhead cost ($\$6 \text{ per machine-hour} \times 15,000 \text{ machine-hours} = \$90,000$) would be applied to Work in Process. The following entry records the application of Manufacturing Overhead to Work in Process:

	(7)	DR	CR	
Work in Process				90,000
Manufacturing Overhead				90,000

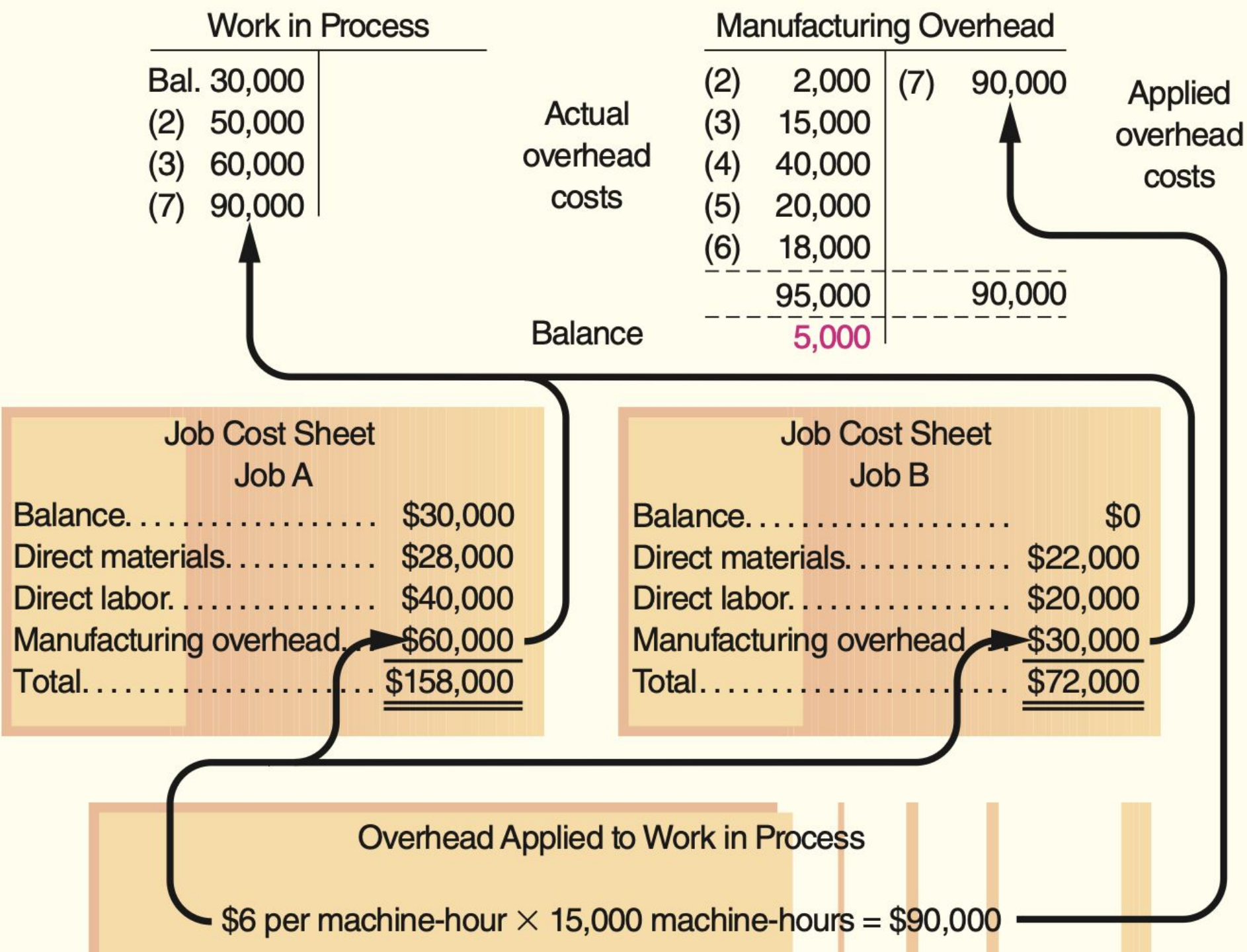
The Concept of a Clearing Account

Manufacturing Overhead (a clearing account)

Actual overhead costs are charged to this account as they are incurred throughout the period.

Overhead is applied to Work in Process using the predetermined overhead rate.

The Flow of Costs in Overhead Application



Nonmanufacturing Costs

To illustrate the correct treatment of nonmanufacturing costs, assume that Ruger Corporation incurred \$30,000 in selling and administrative salary costs during April. The following entry summarizes the accrual of those salaries:

(8)	DR	CR
Salaries Expense		30,000
Salaries and Wages Payable		30,000

Assume that depreciation on office equipment during April was \$7,000. The entry is as follows:

(9)	DR	CR
Depreciation Expense		7,000
Accumulated Depreciation		7,000

Nonmanufacturing Costs

Finally, assume that advertising was \$42,000 and that other selling and administrative expenses in April totaled \$8,000. The following entry records these items:

	(10)	DR	CR
Advertising Expense			42,000
Other Selling and Administrative Expense			8,000
Accounts Payable *			50,000

*Other accounts, such as Cash may be credited.

Cost of Goods Manufactured

In the case of Ruger Corporation, assume that Job A was completed during April. The following entry transfers the cost of Job A from Work in Process to Finished Goods:

	(11)	DR.	CR.
Finished Goods			158,000
Work in Process			158,000

Cost of Goods Sold

For Ruger Corporation, we will assume 750 of the 1,000 gold medallions in Job A were shipped to customers by the end of the month for total sales revenue of \$225,000. Because 1,000 units were produced and the total cost of the job from the job cost sheet was \$158,000, the unit product cost was \$158. The following journal entries would record the sale (all sales were on account):

(12)	DR.	CR.
Accounts Receivable		2,25,000
Sales		
2,25,000		
(13)	DR.	CR.
Cost of Goods Sold		118,500
Finished Goods (750 units * \$158 per unit = \$118,500)		118,500

Schedules of Cost of Goods Manufactured and Cost of Goods Sold

Cost of Goods Manufactured

Direct materials:

Beginning raw materials inventory	\$ 7,000	
Add: Purchases of raw materials	<u>60,000</u>	
Total raw materials available	67,000	
Deduct: Ending raw materials inventory	<u>15,000</u>	
Raw materials used in production	52,000	
Deduct: Indirect materials included in manufacturing overhead	<u>2,000</u>	\$ 50,000
Direct labor		60,000
Manufacturing overhead applied to work in process		<u>90,000</u>
Total manufacturing costs		200,000
Add: Beginning work in process inventory		<u>30,000</u>
		230,000
Deduct: Ending work in process inventory		<u>72,000</u>
Cost of goods manufactured		<u><u>\$158,000</u></u>

Cost of Goods Sold

Beginning finished goods inventory	\$ 10,000
Add: Cost of goods manufactured	<u>158,000</u>
Cost of goods available for sale	168,000
Deduct: Ending finished goods inventory	<u>49,500</u>
Unadjusted cost of goods sold	118,500
Add: Underapplied overhead	<u>5,000</u>
Adjusted cost of goods sold	<u><u>\$123,500</u></u>

Schedules of Cost of Goods Manufactured and Cost of Goods Sold

$$\text{Raw materials used in production} = \text{Beginning raw materials inventory} + \text{Purchases of raw materials} - \text{Ending raw materials inventory}$$

$$\text{Total manufacturing costs} = \text{Direct materials} + \text{Direct labor} + \text{Manufacturing overhead applied to work in process}$$

$$\text{Cost of goods manufactured} = \text{Total manufacturing costs} + \text{Beginning work in process inventory} - \text{Ending work in process inventory}$$

$$\text{Unadjusted cost of goods sold} = \text{Beginning finished goods inventory} + \text{Cost of goods manufactured} - \text{Ending finished goods inventory}$$

Income Statement

Ruger Corporation Income Statement For the Month Ending April 30

Sales		\$225,000
Cost of goods sold (\$118,500 + \$5,000)		<u>123,500</u>
Gross margin		101,500
Selling and administrative expenses:		
Salaries expense	\$30,000	
Depreciation expense	7,000	
Advertising expense	42,000	
Other expense	<u>8,000</u>	<u>87,000</u>
Net operating income		<u><u>\$ 14,500</u></u>

A General Model of Cost Flows

